STPA – Assignment 12 (Birth Death Process – Queueing Theory)

1. A branch office of a large engineering firm has one on-line terminal that is connected to a central computer system during the normal eight-hour working day. Engineers, who work throughout the city, drive to the branch office to use the terminal to make routine calculations. Statistics collected over a period of time indicate that the arrival pattern of people at the branch office to use the terminal has a Poisson (random) distribution, with a mean of 10 people coming to use the terminal each day. The distribution of time spent by an engineer at a terminal is exponential, with a mean of 30 minutes. The branch office receives complains from the staff about the terminal service. It is reported that individuals often wait over an hour to use the terminal and it rarely takes less than an hour and a half in the office to complete a few calculations. The manager is puzzled because the statistics show that the terminal is in use only 5 hours out of 8, on the average. This level of utilization would not seem to justify the acquisition of another terminal. What insight can queuing theory provide?

2. Customers arrive at a sale counter managed by a single person according to poisson process with mean rate of 20 per hour. The time required to serve a customer has an exponential distribution with a mean of 100 seconds. Find the average waiting time of a customer.

3. Customer arrive at a one man barber shop according to a poisson process with a mean inter arrival time of 20 min. Customers spend an average of 15 min. in the barber chair. The service time is exponentially distributed. If an hour is used as a unit of time then,
   (a) What is the probability that a customer need not wait for a hair cut ? (b) What is the expected no. of customers in the barber shop and in the queue?
   (c) How much time can a customer expect to spend in the barber shop? (d) Find the average time that the customers spend in the queue.
   (e) What is the probability that there will be 6 or more customers waiting for service?
   (f) estimate the fraction of the day that the customer will be idle?

4. if people arrive to purchase cinema tickets at the average rate of 6 per minute and it takes 7.5 seconds on the average to purchase a ticket. If a person arrives just 2 minutes before the picture starts and it takes exactly 1.5 minutes to reach the correct seat after purchasing the ticket.
   i) Can he expect to be seated for the start of the picture?
   ii) What is the probability that he will be seated when the film starts?
   iii) How early must he arrive in order to be 99% sure of being seated for the start of the picture?

5. A TV repairman finds that the time spend on his job has an exponential distribution with mean 30 minutes. If he repair sets in the order in which they come in and if the arrival of sets is approximately Poisson with an average rate of 10 per 8 hour day.
   a) Find the repairman’s expected idle time on each day?
   b) How many jobs are ahead of average set just brought?

6. Customers arrive at the express check out lane in a supermarket in a Poisson process with a rate of 15 per hour. The time to check-out a customer is an exponential random variable with' mean of' 2 minutes. Find the average number of customers present. What is the expected waiting time for a customer in the system?

7. Arrivals at a telephone booth are considered to be Poisson with an average time of 12 min. between one arrival and the next. The length of a phone call is assumed to be distributed exponentially with mean 4 min. (a) Find the average number of persons waiting in the system.
   (b) What is the probability that a person arriving at the booth will have to wait in the queue?
   (c) What is the probability that it will take him more than 10 min. altogether to wait for the phone and complete his call?
   (d) Estimate the fraction of the day that the phone will be in use.
   (e) the telephone department will install a second booth, when convinced that an arrival has to wait on the average for atleast 3 minutes for phone. By how much the flow of arrivals should increase in order to justify the second booth?

8. People arrive at a theatre ticket booth in Poisson distributed arrival rate of 25 per hour. Average service time is 2 minutes following exponential distribution. Calculate (1) the mean number in waiting line (2) the mean waiting time (3) the utilization factor (traffic intensity).

9. Assume that at a bank teller window the customer arrives at an average rate of 20 per hour according to Poisson distribution. Service rate of the bank teller is calculated 2 minutes per transaction. Assume also that the bank teller spends a distributed customer who arrive from an infinite population are served on a first come first services basis and there is no limit to possible queue length.

   1. What is the value of utilization factor?
   2. What is the expected waiting time in the system per customer?
   3. What is the probability of zero customer in the system?

10. ABC company has one hob regrinding machine. The hobs needing grinding are sent from company’s tool crib to this machine which is operated one shift per day of 8 hours duration. It takes on the average half an hour to regrind a hob. The arrival of hobs is random with an average of 8 hobs per shift.Calculate the present utilization of hob regrinding machine.What is average time for the hob to be in the regrinding section?

11. The management is prepared to recruit another grinding operator when the utilization of the machine increases to 80%. What should the arrival rate of hobs then be?

12. In a bank with a single server, there are two chairs for waiting customers. On an average one customer arrives every 12 minutes and each customer takes 6 minutes for getting served. Make suitable assumption, find

1. The probability that there are more than two customers in the queue,
2. The probability that there are exactly two customers in the queue
3. Expected waiting time of a customer.

13. At a certain petrol pump, customers arrive according to a Poisson process with an average time of 5 minutes between the arrivals. The service time is exponentially distributed with a mean of 2 minutes based on this information.

    Find out:

    Traffic intensity or utilization

    What would be the average number of customers in the queue?

    What is the expected number of customers at the petrol pump?

    what is the expected time one spends at the petrol pump?

    What would be the proportion of time the petrol pump is idle?